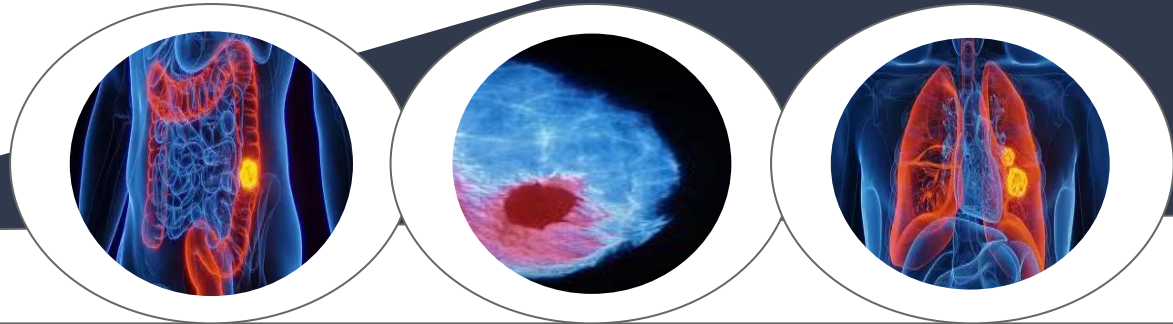


Comparative Analysis of Colorectal, Lung, and Breast Cancers in the United States

By: Ritik Sarraf, Monique Townson, and Daniela Aguilar



Background:



The most **common cancers** are:



Breast



Lung



Colon &
Rectum



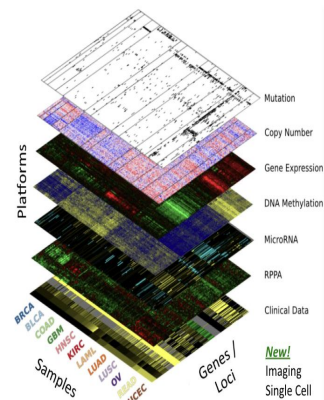
Prostate



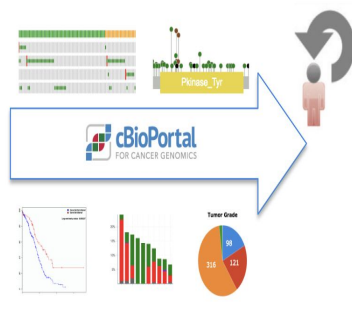
Stomach

- Cancer is the second leading cause of the death in the United States after Heart Disease.
- Breast, colorectal and lung cancers fall under some of the most common diagnosed types
- How do these types of cancers compare?
 - Prevalence
 - Risk factors
 - Survival outcomes
 - Treatment

Introduction



Adapted from TCGA Nat Genet 2014



Biological
discovery

Clinical
interpretation



Lung Cancer

The study titled “*Genomic Features of Response to Combination Immunotherapy in Patients with Advanced Non-Small-Cell Lung Cancer*” from cBioPortal investigates the genetic factors influencing patient responses to combination immunotherapy treatments in advanced non-small-cell lung cancer (NSCLC). By analyzing genomic alterations and tumor mutational burden, this trial provides valuable insights into biomarkers that may predict treatment efficacy, helping guide personalized immunotherapy strategies.

Breast Cancer

The trial “*The Genomic Landscape of Endocrine-Resistant Advanced Breast Cancers*” focuses on identifying genetic changes driving resistance to endocrine therapies in advanced breast cancer patients. Utilizing comprehensive genomic profiling, this study uncovers mutations and alterations associated with therapy resistance, offering potential targets for alternative treatments and improving understanding of tumor evolution under therapy pressure.



Colon Cancer

The “*Colorectal Adenocarcinoma (TCGA, PanCancer Atlas)*” dataset characterizes the molecular and genomic landscape of colorectal cancer through extensive profiling of tumor samples. This resource highlights key mutations, copy number alterations, and pathway disruptions underlying colorectal tumorigenesis, aiding in the classification of tumor subtypes and the identification of prognostic and therapeutic biomarkers.



Datasets:

Cancers Analyzed: Clinical Trials

- **Lung Cancer:**
https://www.cbioportal.org/study/summary?id=nsclc_mskcc_2018
Genomic Features of Response to Combination Immunotherapy in Patients with Advanced Non-Small-Cell Lung Cancer
Focuses on genomic biomarkers predicting response to immunotherapy in advanced NSCLC.
- **Breast Cancer:**
https://www.cbioportal.org/study/summary?id=breast_msk_2018
The Genomic Landscape of Endocrine-Resistant Advanced Breast Cancers
Investigates genetic drivers of resistance to endocrine therapy in advanced breast cancer.
- **Colon Cancer:**
https://www.cbioportal.org/study/summary?id=coadread_tcga_pan_can_atlas_2018
Colorectal Adenocarcinoma (TCGA, PanCancer Atlas)
Comprehensive profiling of molecular alterations in colorectal cancer to inform tumor classification and therapy

Acquired from: <https://www.cbioportal.org/>

Cancer Dataset Description

The dataset contains patient-level clinical and genomic data with the following variables:

- **Patient_ID:** Unique identifier for each patient.
- **Cancer_Type:** Diagnosed cancer type (e.g., Breast, Colorectal, Lung).
- **Diagnosis_Age:** Age at cancer diagnosis.
- **Sex:** Biological sex (Male or Female).
- **Stage:** Clinical stage at diagnosis, indicating tumor progression.
- **Overall_Survival_Months:** Survival duration (months) since diagnosis or treatment start.
- **Overall_Survival_Status:** Patient status at last follow-up (alive or deceased).
- **Progression_Free_Months:** Time patient remained free from disease progression (months).
- **Progression_Free_Status:** Indicates if disease progression has occurred.
- **Mutation_Count:** Total number of somatic mutations detected in the tumor.
- **Fraction_Genome_Altered:** Proportion of genome affected by copy number alterations.
- **TMB (Tumor Mutational Burden):** Number of mutations per megabase of DNA, indicating mutation load.

Additional Dataset:

“AH Provisional Cancer Death Counts by Month and Year, 2020-2021”

Acquired from: [Data.gov](https://data.gov)

Description: provisional death counts of different cancers by month, year, and other demographics using death certificates for U.S. residents during the years 2020-2021

Count: 8,448 rows of observations and 11 variables

Variables:

Actual Date: Date as of report was created/published

Start Date: Indicates initial date of data collection for the period

End Date: Indicates final date of data collection for the period

Country: Country in which data is collected (United States)

Year: Year of data collection (2020-2021)

Month: Month of data collection

Sex: Biological sex (Female or Male)

Age Group: Years (0-14, 15-24, 25-34, 35-44, 45-54, 55-64, 65-74, 75 and over)

Race and Hispanic Origin: Self-identified race/ethnicity group

Cancer Types: Diagnosed cancer types

Number of Deaths: Total number of deaths recorded

Data Cleaning



Merging: The three individual datasets pertaining to lung, breast, and colorectal cancers were merged by common variables (Patient ID, Cancer Type, Diagnosis Age, Sex, Stage, Overall Survival Months, Overall Survival Status, Progression Free Months, Progression Free Status, Mutation Count, Fraction Genome Altered and TMB)

Reshaping: The dataset that wasn't merged was reshaped to long format to consolidate cancer types into one variable, and total deaths as its corresponding value

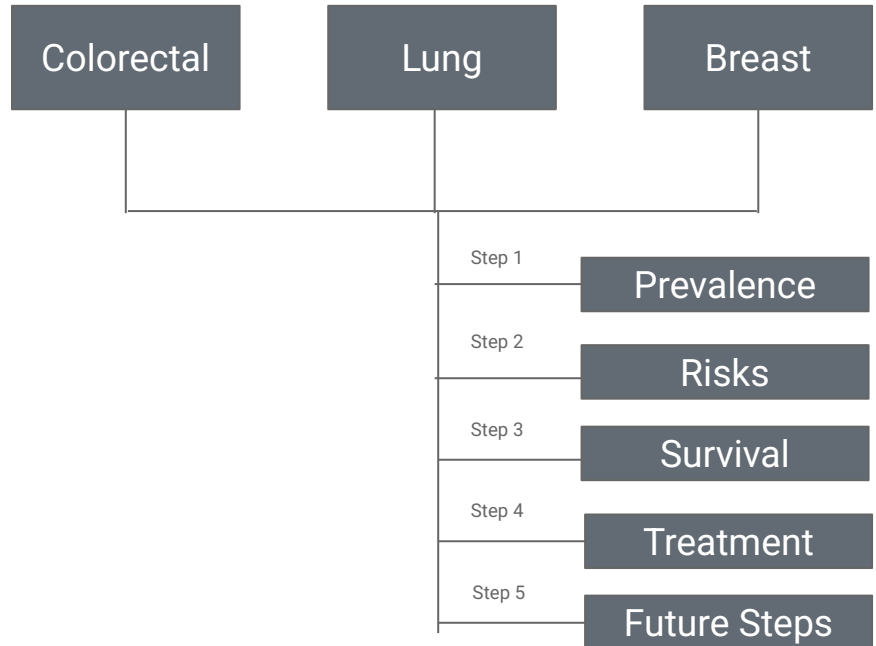
NA's: NA's found in variables used for analysis were removed in R and cleaned merged dataset were used.

Standardization of variables: Completed in Excel to ensure consistency of entries especially within the categorical variables

Research Question:



How do colorectal, lung, and breast cancers compare in prevalence, risk factors, survival outcomes, and responsiveness to treatment in the United States?



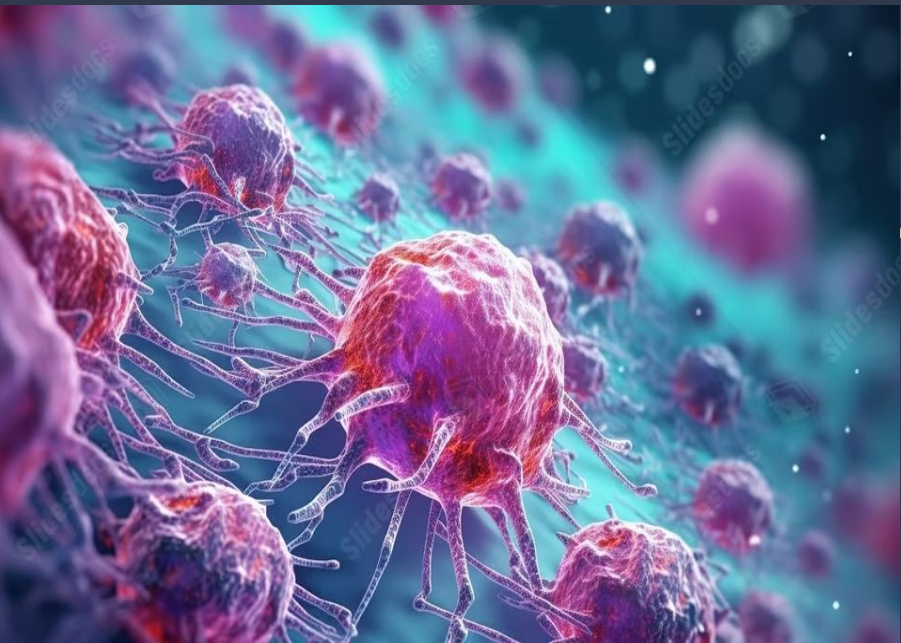
Summary Statistics:

Merged Dataset

Merged Dataset

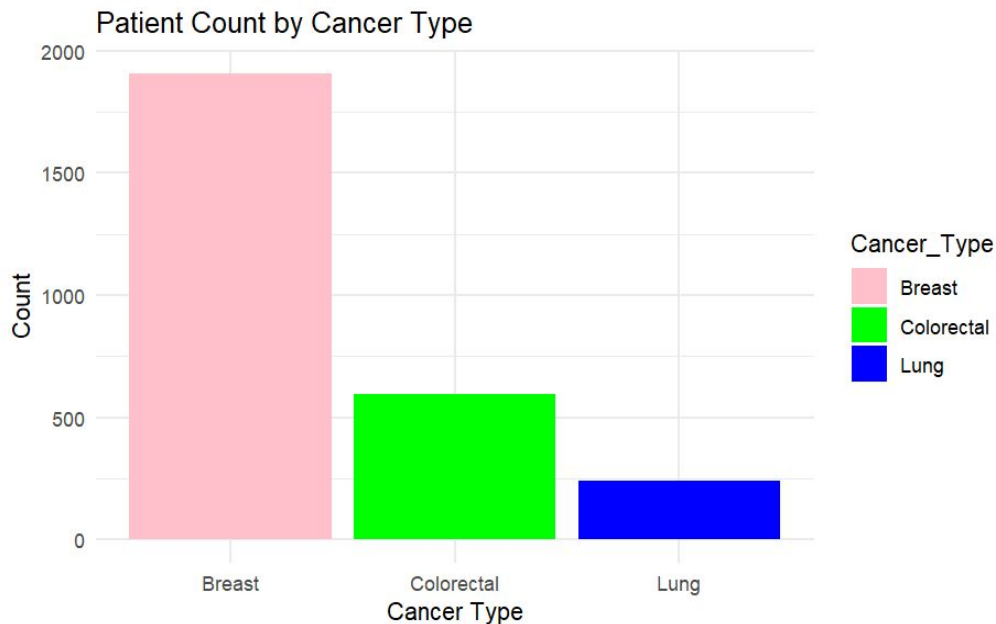
- It contains 2598 rows and 12 columns and the diagnosis age ranging from 22 to 92 years.
- More than 50 percent of the diagnosed patients are above 50 years of age.
- Overall Mean Survival Months ~ 58 months
- Mean Progression Free Months ~ 33 months
- Mean Mutation Count ~ 85
- Mean Fraction Genome Altered ~ 0.295
- Mean TMB ~ 3.87

Prevalence Data of Breast, Colorectal, & Lung Cancer

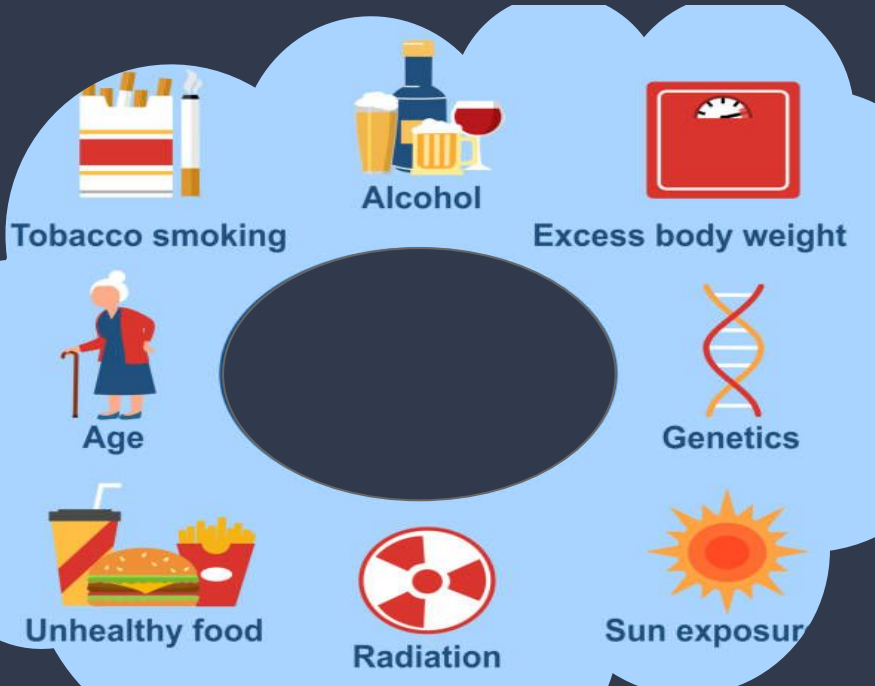


The graph shows how many patients were included for each cancer type. Breast cancer had the most patients, followed by colorectal cancer, and lung cancer with the fewest.

- **Lung Cancer Study:** Looked at how genetic differences affect whether patients respond well to new immunotherapy treatments.
- **Breast Cancer Study:** Focused on why some breast cancers stop responding to hormone-based treatments and explore new options.
- **Colorectal Cancer Study:** Research highlights key changes in colorectal tumors that may guide better diagnosis and treatment choices.

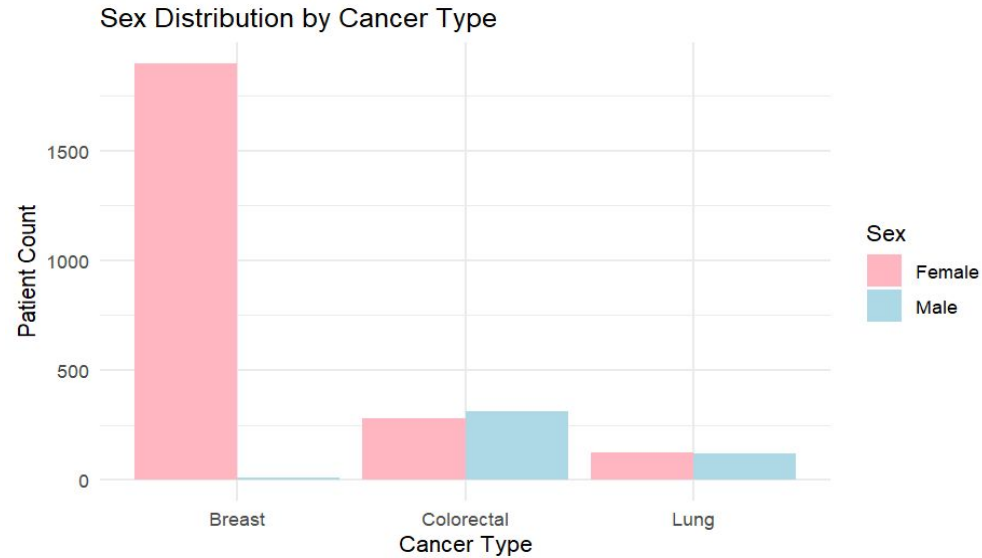


Risk Factors for Developing Cancer:



- Age
- Sex
- Race
- Genetics
- Smoking
- Obesity
- Alcohol use
- Diet
- Hormonal factors, etc.

Patients Gender



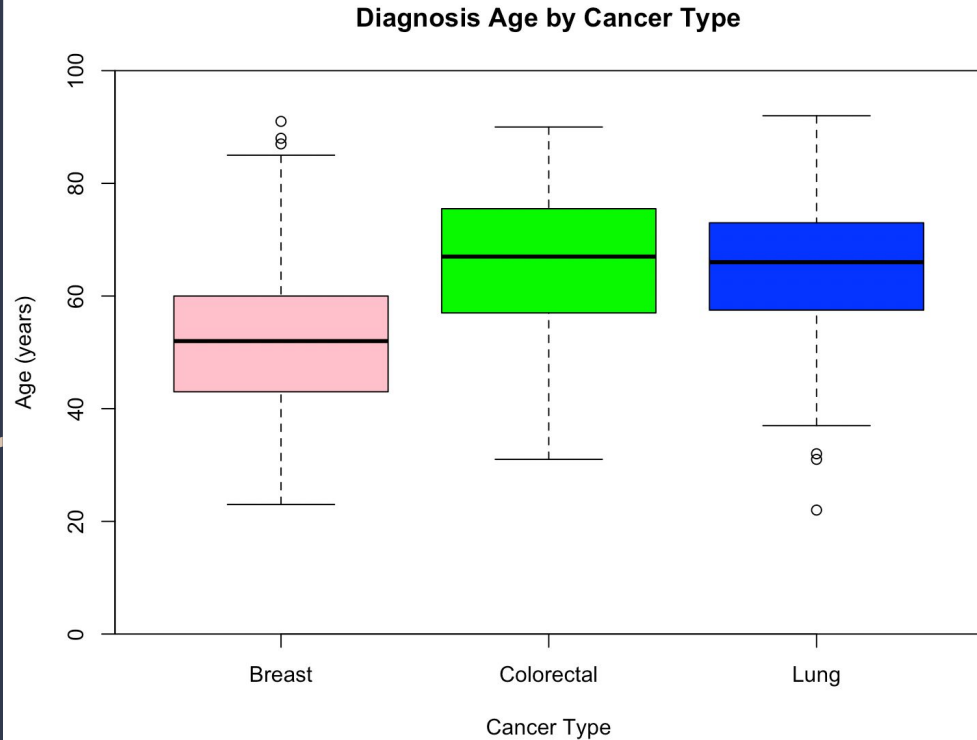
This bar chart illustrates the distribution of patients by sex across three cancer types: Breast, Colorectal, and Lung cancer.

- **Breast cancer** patients are predominantly female, with a significantly higher number of females compared to males, reflecting known epidemiology.
- **Colorectal cancer** shows a more balanced distribution between male and female patients, with slightly more males.
- **Lung cancer** has a higher number of male patients compared to females.

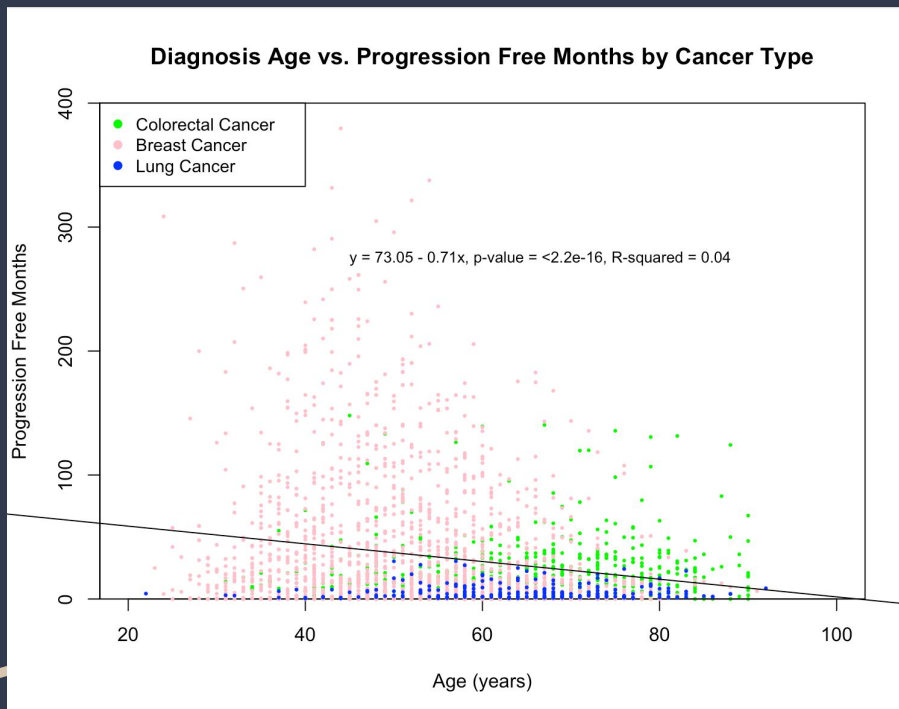
This gender distribution highlights important demographic differences among these cancers, which can influence disease presentation, treatment response, and outcomes.

Risk Factors – Age:

- Age is one of the more significant risk factors of cancer:
 - Diagnosis age medians are centered around middle ages and older
- Breast cancer shows earlier onset



Risk Factors – Age:

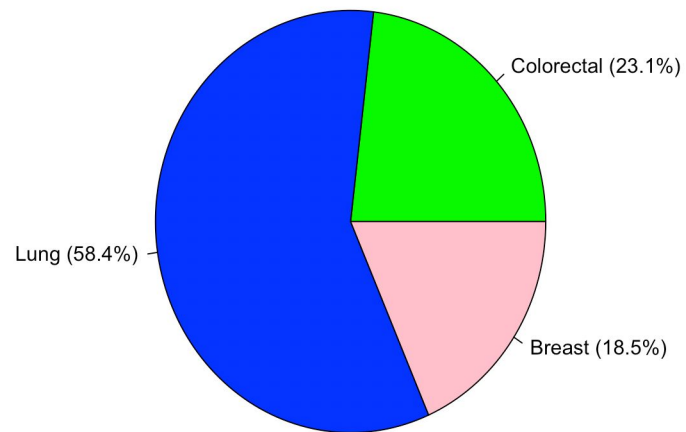


- As age increases, the number of progression-free months decrease.
 - Higher chance of cancer progression the older you get
 - The correlation for this regression is -0.20, therefore the relationship between progression free months and age is not very strong
 - Even though the p-value is less than 0.05, the R-squared is also very low at 0.04
- Breast cancer:
 - Least progression, especially in people under 60 years
 - Affects younger individuals
- Lung cancer:
 - Highest chance of progression
 - Most present in older individuals
- Colorectal cancer:
 - Progression is not as low as breast cancer patients but not as high as lung cancer patients
 - Affects mainly people age 60+

Survival Outcomes – Cancer Type:

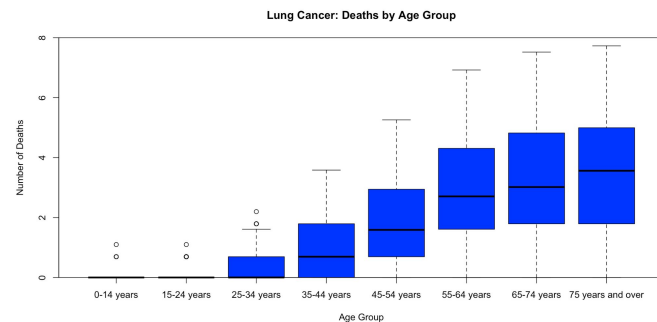
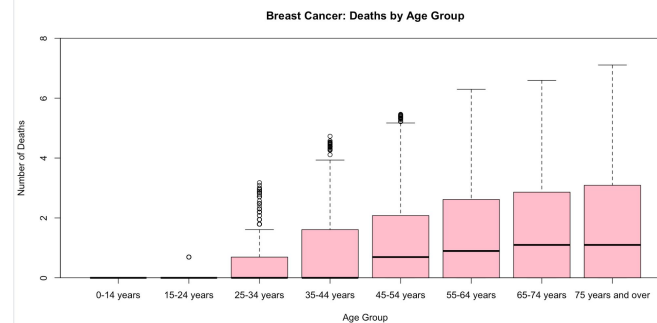
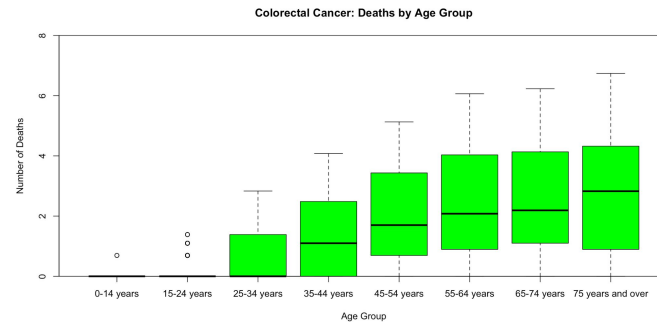
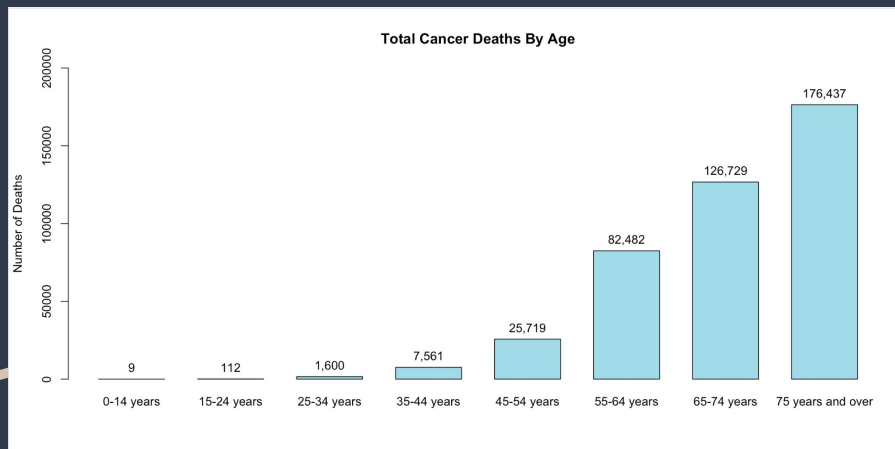
- Pie chart shows proportion of deaths in U.S. (years 2020-2021) for each type of cancer
 - Lung: 245,783 ~ (58.4%)
 - Colorectal: 97,253 ~ (23.1%)
 - Breast: 77,613 ~ (18.5%)

Proportion of Deaths by Cancer Type

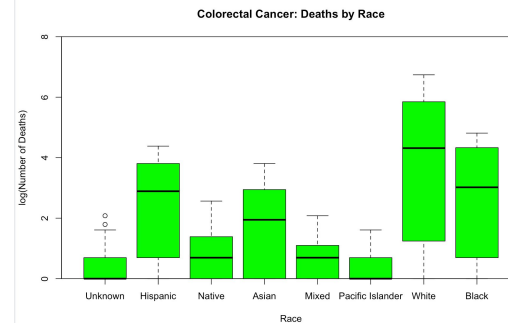
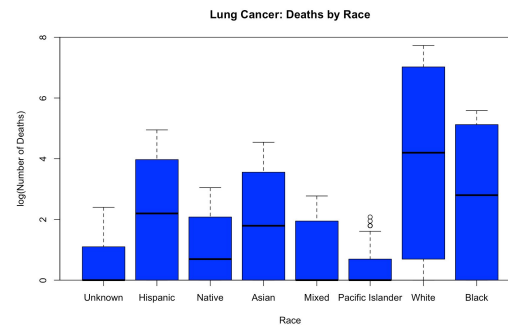
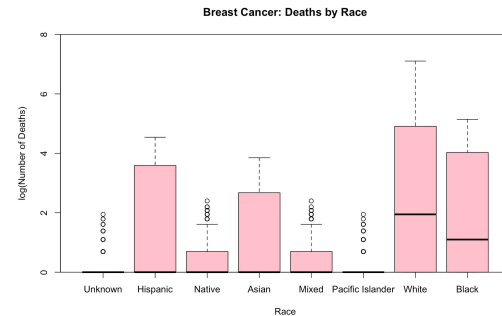
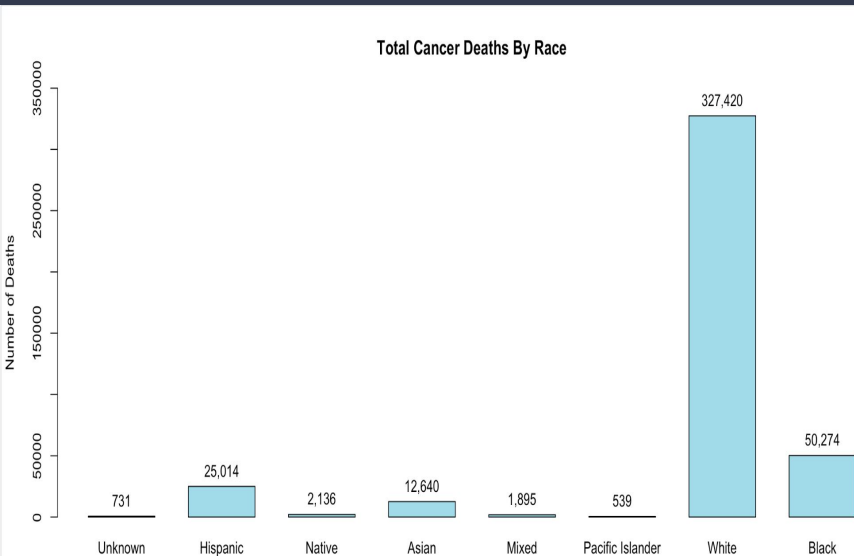


Survival Outcomes – Age:

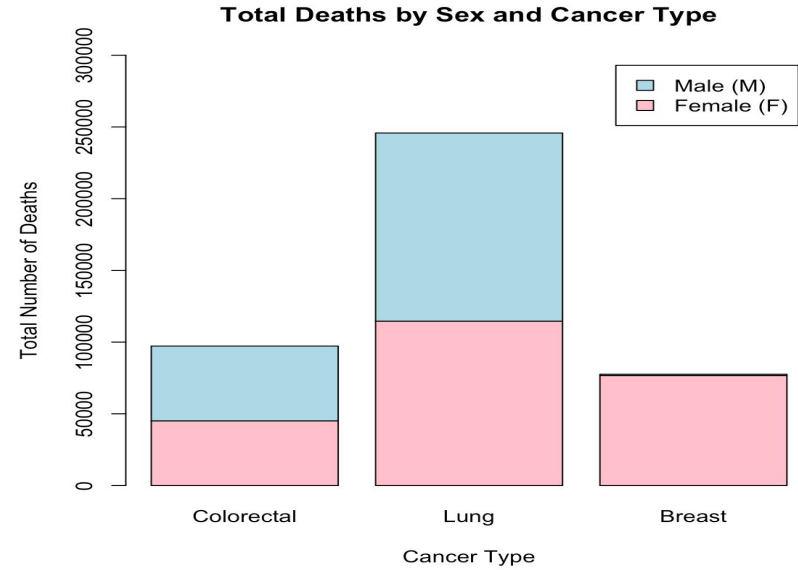
- Older ages coincide with higher number of deaths across all cancer types



Survival Outcomes – Race:



Survival Outcomes – Sex:



Across cancer types, **males generally experience higher mortality** than females, especially for lung and colorectal cancers. This may point to differences in risk factors, early detection, or treatment outcomes by sex.

- **Lung cancer** accounts for the highest number of deaths among both males and females, with males experiencing **notably higher mortality** than females.
- **Colorectal cancer** deaths are more balanced between sexes but still show slightly **higher male mortality**.
- **Breast cancer deaths** occur almost exclusively in **females**, as expected due to biological prevalence.



Mutations Across Cancer Types:

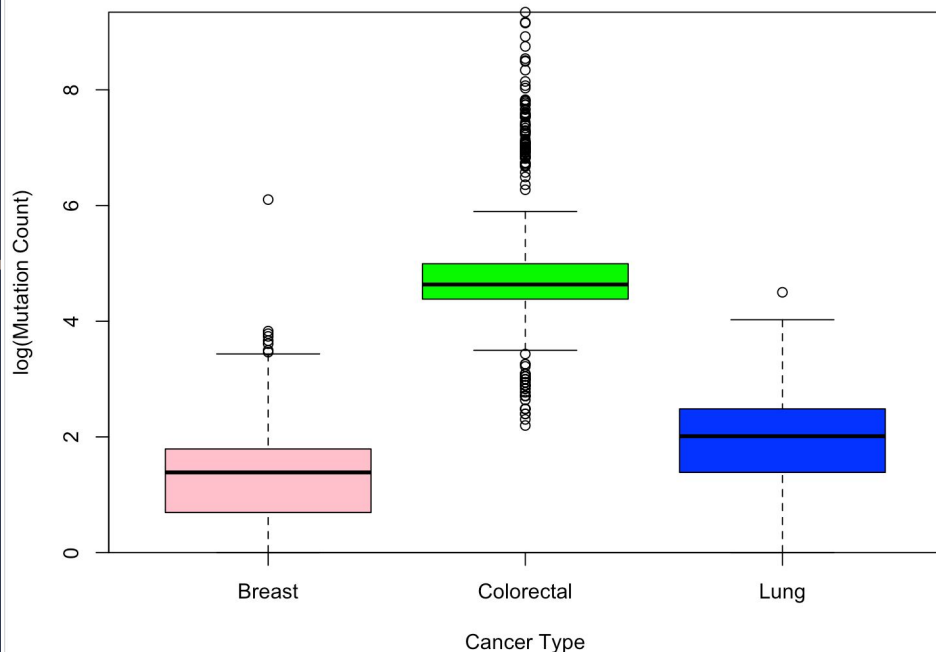
Cancer types vary significantly in their mutation profiles, which may influence treatment strategies, colorectal cancers may be more responsive to mutation-targeted therapies.

Mutations can be caused by:

- Genetics
- Environmental factors
- Random error

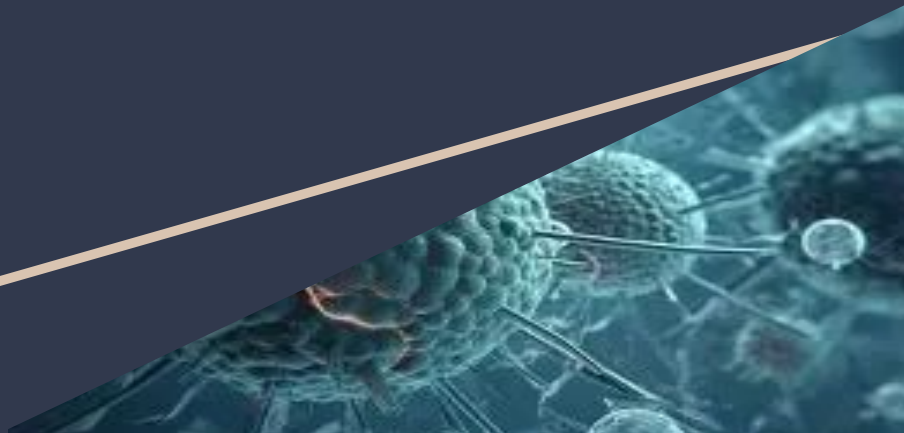
- **Colorectal cancer** shows the **highest mutation counts**, with a wide range and many outliers, indicating high genetic variability.
- **Lung cancer** has **moderate mutation counts**, likely influenced by environmental exposures like smoking or pollution.
- **Breast cancer** displays the **lowest mutation burden**, with a tighter range and fewer extreme values.

Mutation Count by Cancer Type



Predicting Responsiveness to Treatment Using Tumor Mutational Burden (TMB):

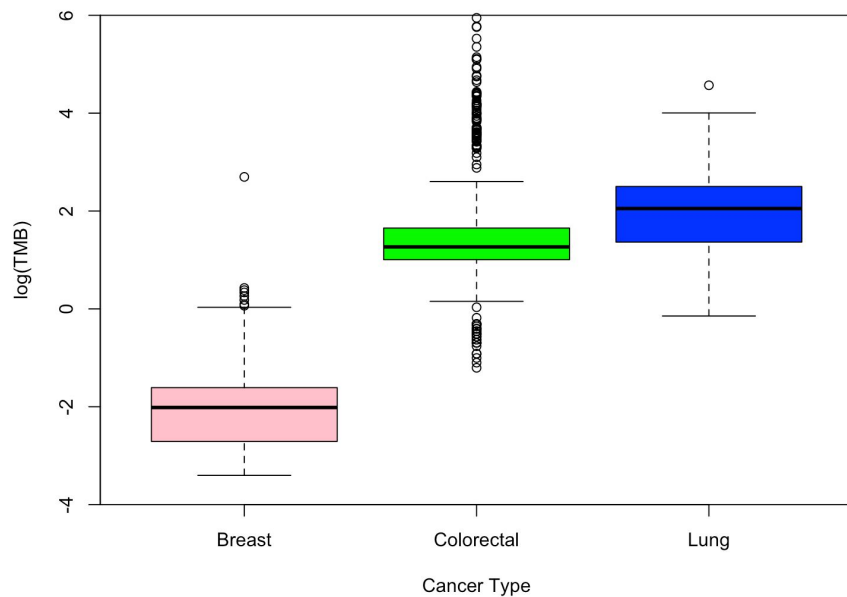
- TMB: # of mutations within a cancer cell's DNA (mutations per megabase(mut/Mb))
- TMB is standardized and better used for comparison whereas mutation counts show counts without relation to the size of the region of DNA
- Used to predict response to immunotherapy treatments (i.e. higher mutations may suggest a better response)
- Higher TMB predicts higher response. Doesn't necessarily mean higher survival or better outcome



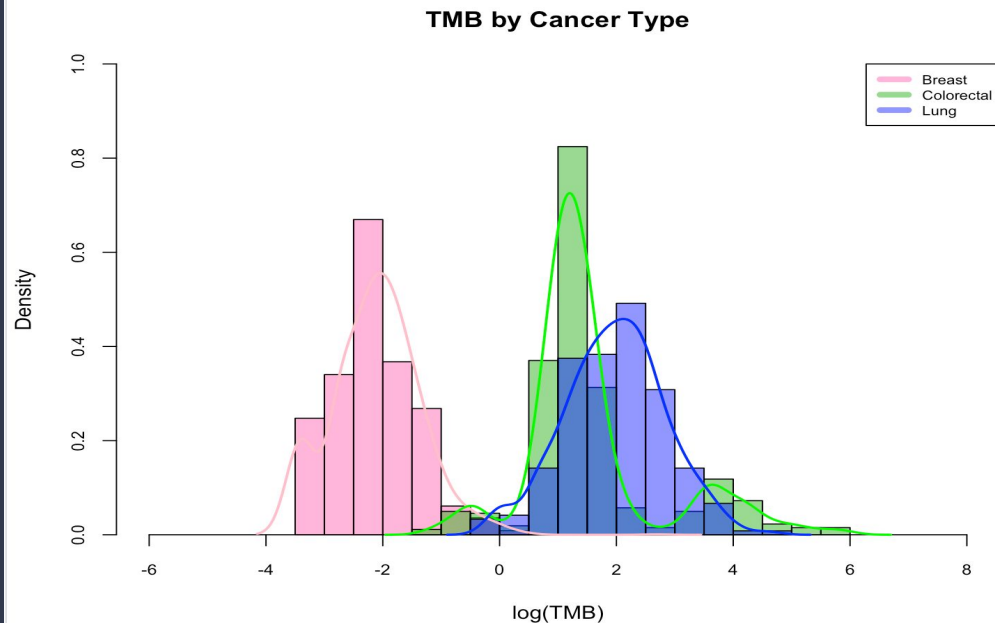
TMB by Cancer Type:

- In comparison, **lung cancer** seems to have the **largest median log(TMB)** of about **2**.
- **Colorectal Cancer** lies just below the lung cancer with the median log(TMB) of around **1.7 with smallest IQR**. This shows that the log(TMB) is more consistent with Colorectal cancer type.
- Breast cancer had the median log(TMB) of -2.

Tumor Mutational Burden by Cancer Type



TMB Across Cancer Types:



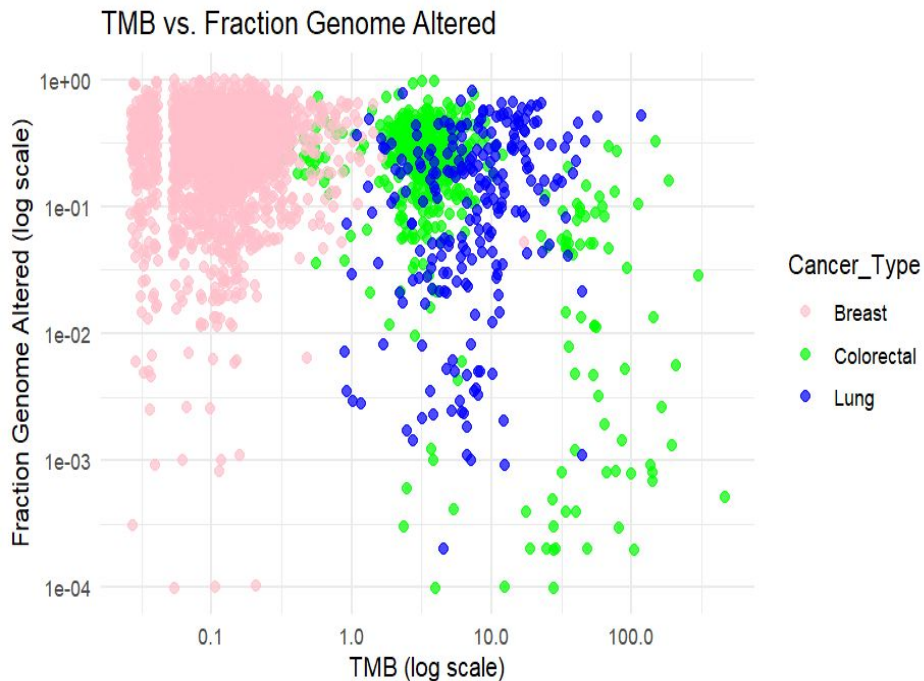
This density plot shows the distribution of **log-transformed Tumor Mutational Burden (TMB)** across three cancer types:

- **Breast cancer** (pink) has the **lowest TMB values**, with most cases centered around $\log(\text{TMB}) \approx -2$.
- **Colorectal cancer** (green) shows a **narrower and more consistent** distribution, with a peak near $\log(\text{TMB}) \approx 1$.
- **Lung cancer** (blue) exhibits the **highest TMB values** and a wider distribution, indicating greater variability in mutation burden.

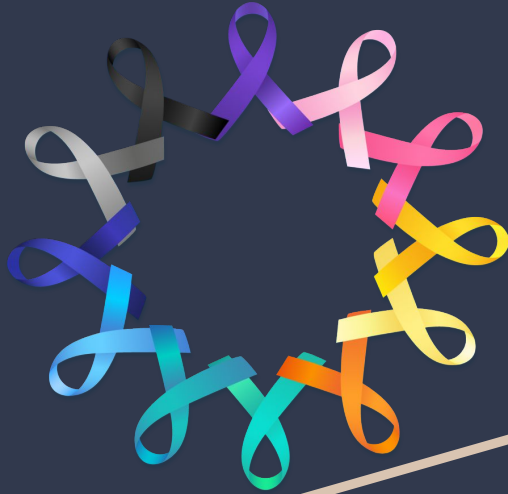
Changes in Cancer DNA

This graph shows two important measures of how much a cancer's DNA has changed: **Tumor Mutational Burden (TMB)** – how many mutations are found in the tumor's DNA. **Fraction Genome Altered** – how much of the tumor's DNA has larger changes, like missing or extra pieces.

- **Breast cancers (pink)** generally have lower mutation and alteration levels.
- **Colorectal cancers (green)** show more variation, with some tumors having higher DNA changes.
- **Lung cancers (blue)** tend to have the highest mutation burden and genome changes, which can make them more responsive to immunotherapy (based on predictive markers)



Conclusion:



- Cancer can be considered an age-related disease
 - As age increases, so does the prevalence of cancer in the population
- Breast cancer is most present in females. However, few males can also develop the disease.
- Breast cancer is the most prevalent of the three, and it starts to show up at earlier ages
- Lung cancer is the deadliest of the three. It shows the largest proportion of deaths even though it isn't the most prevalent
- People who are white, black, or hispanic show the highest number of deaths across all cancers
- Lung cancer shows the highest numbers of TMB, indicating it may be more responsive to treatments such as immunotherapies based on predictive factors

Future Steps

When analyzing our data, it seems like age and genetics play a vital role in the growth and progression of cancer cells.

However, it might also be important to focus on the lifestyle as it has deep correlation with cancer types.

Future steps could include researching the effects of the following risk factors on these cancer types:

1. Obesity (BMI)
2. Diet
3. Smoking
4. Alcohol Consumption



Source/References

- [Getty Images](#)

Datasets:

- https://www.cbiportal.org/study/summary?id=nsclc_mskcc_2018
- https://www.cbiportal.org/study/summary?id=breast_msk_2018
- https://www.cbiportal.org/study/summary?id=coadread_tcg_a_pan_can_atlas_2018
- <https://catalog.data.gov/dataset/ah-provisional-cancer-death-counts-by-month-and-year-2020-2021-ab4a5>

Additional information:

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- Mary C. White, Dawn M. Holman, Jennifer E. Boehm, Lucy A. Peipins, Melissa Grossman, S. Jane Henley, Age and Cancer Risk: A Potentially Modifiable Relationship, *American Journal of Preventive Medicine*, Volume 46, Issue 3, Supplement 1, 2014, Pages S7-S15, ISSN 0749-3797
- The Use of Breast Imaging to Screen Women at High Risk for Cancer Sickles, Edward A. *Radiologic Clinics*, Volume 48, Issue 5, 859 - 878
- Lung Cancer Epidemiology, Risk Factors, and Prevention de Groot, Patricia et al. *Radiologic Clinics*, Volume 50, Issue 5, 863 - 876; doi: 10.1016/j.rcl.2012.06.006

Thank you! 😊

Any Questions?